

Data used: (data-preprocessing-pipeline.ipynb)

TARGET =====

Microbiology phenotype (resistant or not)

FEATURES =====

Area Deprivation Index Comorbidities - binary flag for each possible comorbidity Age Gender Last Nursing Home Visit Prior Antibiotic Used Corresponding to Order - this ensures we are looking at the positive test if it exists Time Since Antibiotic Used Corresponding to Order Most recent procedure Time since most recent procedure Inpatient? Outpatient? ER? Urgent Care? Day surgery?

From this data, we can get the following metrics: (classical_ml_data_validation.ipynb) Accuracy: 0.8137959939998235 Confusion Matrix True negatives: 33472 (0.7383746580781788) False positives: 1592 (0.03511867996117533) False negatives: 6849 (0.15108532603900116) True positives: 3419 (0.07542133592164475)

22.65% Positive occurrence rate

DATA PREPROCESSING =====

Removed rows for which subjects had no listed drug code (unlikely to be useful data anyway) Reduced several ward flags to a single integer encoding PCA to 11 features on comorbidities preserves 80.57402429560702% of information. To 15 features: 81.9202600991625%-npf 81.9202503870941% To 21 features: 83.59505954298974%

CML RESULTS =====

HistGradientBoostingClassifier: learning_rate=0.01, max_iter=5000 Accuracy: 72.77199329392041% Confusion Matrix True negatives: 25428 (0.5609282625959587) False positives: 9636 (0.2125650754433954) False negatives: 2707 (0.059714991617400515) True positives: 7561 (0.16679167034324538) AUROC: 0.8044 PRAUC: 0.5768

hgbc_alpha0.01_maxiter5000_521features Accuracy: 73.4205417806406% Confusion Matrix True negatives: 25540 (0.5633989234977499) False positives: 9524 (0.21009441454160416) False negatives: 2525 (0.05570016765198976) True positives: 7743 (0.17080649430865613) AUROC: 0.8167 PRAUC: 0.5950

hgbc_alpha0.01_maxiter5000_24features_npf Accuracy: 72.7036089296744% True negatives: 25527 (0.5631121503573635) False positives: 9537 (0.21038118768199066) False negatives: 2837 (0.06258272302126533) True positives: 7431 (0.16392393893938056) AUROC: 0.8004 PRAUC: 0.5697

hgbc_alpha0.01_maxiter5000_24features Accuracy: 72.79184681902409% Confusion Matrix True negatives: 25475 (0.5619650577958175) False positives: 9589 (0.21152828024353656) False negatives: 2745 (0.060553251566222537) True positives: 7523 (0.16595341039442338) AUROC: 0.8049 PRAUC: 0.5744

hgbc_alpha0.01_maxiter5000_521features_undersampled Accuracy: 72.74331597988176% Confusion Matrix True negatives: 25305 (0.5582149474984558) False positives: 9759 (0.21527839054089826) False negatives: 2597 (0.057288449660284124) True positives: 7671 (0.16921821230036177) AUROC: 0.8112 PRAUC: 0.5848

hgbc_alpha0.01_maxiter5000_521features_undersampled_5000samples Accuracy: 0.738 Confusion Matrix True negatives: 653 (0.653) False positives: 121 (0.121) False negatives: 141 (0.141) True positives: 85 (0.085) AUROC: 0.6892 PRAUC: 0.4066

hgbc_alpha0.01_maxiter5000_521features_undersampled_10000samples Accuracy: 0.752 Confusion

Matrix True negatives: 1262 (0.631) False positives: 285 (0.1425) False negatives: 211 (0.1055) True positives: 242 (0.121) AUROC: 0.7421 PRAUC: 0.4755

RESULTS BY ANTIBIOTIC ==

CIP =

CIP_hgbc_alpha0.01_maxiter5000_521features Accuracy: 0.894447223611806 Confusion Matrix True negatives: 1170 (0.5852926463231616) False positives: 120 (0.060030015007503754) False negatives: 91 (0.04552276138069034) True positives: 618 (0.3091545772886443) AUROC: 0.9576 PRAUC: 0.9329

CIP_hgbc_alpha0.01_maxiter5000_521features_undersampled Accuracy: 0.7078539269634817 Confusion Matrix True negatives: 957 (0.47873936968484243) False positives: 333 (0.16658329164582292) False negatives: 251 (0.12556278139069535) True positives: 458 (0.22911455727863933) AUROC: 0.7631 PRAUC: 0.6428

CIP_hgbc_alpha0.01_maxiter5000_521features_oversampled Accuracy: 0.6948474237118559 Confusion Matrix True negatives: 843 (0.42171085542771386) False positives: 447 (0.2236118059029515) False negatives: 163 (0.0815407703851926) True positives: 546 (0.27313656828414207) AUROC: 0.7754 PRAUC: 0.6169

CRO =

CRO_hgbc_alpha0.01_maxiter5000_521features Accuracy: 0.7668531841193712 Confusion Matrix True negatives: 2084 (0.555289102051692) False positives: 690 (0.1838529176658673) False negatives: 185 (0.049293898214761524) True positives: 794 (0.21156408206767918) AUROC: 0.8626 PRAUC: 0.7128 CRO_hgbc_alpha0.01_maxiter5000_521features_oversampled Accuracy: 0.36504130029309884 Confusion Matrix True negatives: 485 (0.12922994937383428) False positives: 2289 (0.609912070343725) False negatives: 94 (0.025046629363176125) True positives: 885 (0.23581135091926458) AUROC: 0.6480 PRAUC: 0.4795

CRO_hgbc_alpha0.01_maxiter5000_521features_undersampled Accuracy: 0.7921662669864109 Confusion Matrix True negatives: 2108 (0.5616839861444178) False positives: 666 (0.1774580335731415) False negatives: 114 (0.030375699440447643) True positives: 865 (0.23048228084199307) AUROC: 0.8970 PRAUC: 0.7553

CFZ =

CFZ_hgbc_alpha0.01_maxiter5000_521features Accuracy: 0.782432010003126 Confusion Matrix True negatives: 1974 (0.6170678336980306) False positives: 549 (0.1716161300406377) False negatives: 147 (0.045951859956236324) True positives: 529 (0.16536417630509534) AUROC: 0.8685 PRAUC: 0.6927

CFZ_hgbc_alpha0.01_maxiter5000_521features_oversampled Accuracy: 0.5270396999062207 Confusion Matrix True negatives: 1073 (0.3354173179118475) False positives: 1450 (0.45326664582682086) False negatives: 63 (0.019693654266958426) True positives: 613 (0.19162238199437323) AUROC: 0.8044 PRAUC: 0.5498

CFZ_hgbc_alpha0.01_maxiter5000_521features_undersampled Accuracy: 0.7896217567989997 Confusion Matrix True negatives: 1923 (0.6011253516723977) False positives: 600 (0.1875586120662707) False negatives: 73 (0.022819631134729602) True positives: 603 (0.18849640512660207) AUROC: 0.9043 PRAUC: 0.7128

CFZ_hgbc_alpha0.01_maxiter5000_11features Accuracy: 0.7461706783369803 Confusion Matrix True negatives: 1894 (0.5920600187558612) False positives: 629 (0.19662394498280714) False negatives: 183 (0.05720537668021257) True positives: 493 (0.1541106595811191) AUROC: 0.8248 PRAUC: 0.5908

CFZ_hgbc_alpha0.01_maxiter5000_11features_undersampled Accuracy: 0.22006877149109097 Confusion Matrix True negatives: 28 (0.0087527352297593) False positives: 2495 (0.7799312285089091) False negatives: 0 (0.0) True positives: 676 (0.21131603626133166) AUROC:

0.6385 PRAUC: 0.3195 CFZ_hgbc_alpha0.01_maxiter5000_11features_oversampled Accuracy: 0.5792435135979994 Confusion Matrix True negatives: 1609 (0.5029696780243826) False positives: 914 (0.2857142857142857) False negatives: 432 (0.13504220068771491) True positives: 244 (0.07627383557361675) AUROC: 0.5009 PRAUC: 0.2053

AZM =

AZM_hgbc_alpha0.01_maxiter5000_521features Accuracy: 0.9110499770747363 Confusion Matrix True negatives: 1566 (0.7180192572214581) False positives: 146 (0.06694176983035305) False negatives: 48 (0.02200825309491059) True positives: 421 (0.19303071985327833) AUROC: 0.9598 PRAUC: 0.9007 AZM_hgbc_alpha0.01_maxiter5000_521features_oversampled Accuracy: 0.7615772581384685 Confusion Matrix True negatives: 1385 (0.6350298028427327) False positives: 327 (0.1499312242090784) False negatives: 193 (0.088491517652453) True positives: 276 (0.1265474552957359) AUROC: 0.7939 PRAUC: 0.5063 AZM_hgbc_alpha0.01_maxiter5000_521features_undersampled Accuracy: 0.7441540577716643 Confusion Matrix True negatives: 1224 (0.5612104539202201) False positives: 488 (0.22375057313159102) False negatives: 70 (0.03209536909674461) True positives: 399 (0.1829436038514443) AUROC: 0.8634 PRAUC: 0.6420

QML RESULTS =====

Feature reduction (PCA) to 512 features after encoding NA as -99999 (out of range for all features) Train on subset of data for time considerations (order of 1000 subjects) - can run training on multiple subsets if necessary with hot starts Efficient SU2 ansatz results statistics: Quantum VQC on the training dataset: 0.74 Quantum VQC on the test dataset: 0.73 Confusion Matrix: True negatives: 726 (0.726) False positives: 48 (0.048) False negatives: 220 (0.22) True positives: 6 (0.006) Real Amplitudes ansatz results statistics: Quantum VQC on the training dataset: 0.76 Quantum VQC on the test dataset: 0.75

20features_5000samples_efficientsu2map_realamplitudes3reps Note: 2 training sessions (10000 total samples) Test accuracy: 0.7634 Train accuracy: 0.758825 Confusion Matrix True negatives: 7066 (0.7066) False positives: 669 (0.0669) False negatives: 1697 (0.1697) True positives: 568 (0.0568) AUROC: 0.6160 PRAUC: 0.3515

30features_efficientsu2map_realamplitudes3reps Note: used 3 reps on efficientsu2 map - I think this was the problem for us Train Accuracy: 0.69 Test accuracy: 0.69 Confusion Matrix True negatives: 6301 (0.6301) False positives: 1434 (0.1434) False negatives: 1633 (0.1633) True positives: 632 (0.0632) AUROC: 0.5938 PRAUC: 0.2878

20features_efficientsu2map_realamplitudes3reps_npf Note: 2 training sessions (10000 total samples) Test accuracy: 0.7545 Train accuracy: 0.7582 Confusion Matrix True negatives: 7003 (0.7003) False positives: 732 (0.0732) False negatives: 1723 (0.1723) True positives: 542 (0.0542) AUROC: 0.6070 PRAUC: 0.3465

24features_efficientsu2map_realamplitudes3reps_undersampled Test accuracy: 0.504 Train accuracy: 0.51135 Confusion Matrix True negatives: 3917 (0.3917) False positives: 3818 (0.3818) False negatives: 1142 (0.1142) True positives: 1123 (0.1123) AUROC: 0.5050 PRAUC: 0.2336

20features_efficientsu2map_realamplitudes3reps_undersampled Notes: Undersampling to balance the dataset completely killed its performance on normal data. While this wasn't as big a problem for CML, it was a huge problem for QML. Perhaps this made it so the model could not generalize? Test accuracy: 0.4829 Train accuracy: 0.48075 Confusion Matrix True negatives: 3750 (0.375) False positives: 3985 (0.3985) False negatives: 1186 (0.1186) True positives: 1079 (0.1079) AUROC: 0.4745 PRAUC: 0.2124

NOTES =====

I tried using Aer's sampler, but the results were terrible. Noise issues? Unlikely to be the Aer sampler, problem persists for statevectorsampler. Why?

TODO =====

Test Deep Angle encoding Worked! Use AUROC & PRAUC to get better metrics for QML models Worked! Should we switch from trying to see if we can get a really good model to comparing performance? - Yes Switch over to many models, each for one of the antibiotics? - Yes Combine with above to adjust the research question to as follows: "Are there antibiotics for which classical models outperform quantum models, and in what ways?" - Yes SHAP values to determine features of importance? - Yes SMOTE - Try first Very poor performance on per antibiotic set, even after retraining for 100 more iterations. Will try various options. quantum random forest model? try different feature maps, ansatz try more qubits on the big machine? Make figure showing data preprocessing, model training, results processing Why choose these features?

ATTENDANCE LOG =====

21 Jul (Tu) - 12:14-18:20 22 Jul (We) - 09:28-15:35 23 Jul (Th) - 10:20-16:21 28 Jul (Tu) - 09:31-15:35 29 Jul (We) - 09:52-15:50 30 Jul (Th) - 10:50-16:51 04 Aug (Tu) - 10:15-16:45 05 Aug (We) - 10:20-16:40 06 Aug (Th) - 10:30-15:20 10 Aug (Mo) - QC workshop @ NTU 11 Aug (Tu) - QC workshop @ NTU 12 Aug (We) - 09:50-